

ing out the circuit components is a task common to all branches of electronics. The speciality in VLSI layout is the number of possible ways available to do this; there can be multiple layers of different materials on the same silicon, different arrangements of smaller parts for the same component and so on. Each one has a separate layout."

Layout is responsible for the functionality of the circuit. Power dissipation and speed in a circuit present a trade-off; if we try to optimise one, the other is affected. The choice between the two is determined by the way you choose to layout the circuit components. Layout can also affect the fabrication of VLSI chips, making it either easy or difficult to implement the components on the silicon.

How to tackle the reality?

Know where to apply your skillset. Analogue designs are mostly used for small-transistor-count precision

circuits such as amplifiers, data converters, filters, phase-locked loops and sensors. In digital design, the progress in the fabrication of ICs has enabled designers to create fast and powerful circuits in smaller and smaller devices. This also means that you can pack a lot of functionality into the same area. The biggest application of this ability is in the design of ASICs. These ICs are created for specific purposes. The most common application area for an ASIC is DSP—signal filtering, image compression, etc. To understand better, consider the fact that a digital wrist-watch normally consists of a single IC doing all the time-keeping as well as extra features like games, calendar, etc. SoCs (systems on a chip), on the other hand, are highly complex mixed-signal circuits (digital and analogue on the same chip). Network processor chips and wireless radio chips are examples of an SoC.

If it looks complicated to you,

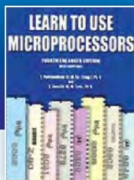
here's an advice: "One can understand this field from chip to ship level only by completing a successful project," says Sasisekar K., president, RCS Online. He advises to select designs that can address both the cost sensitivity and needs of the Indian market; for example, small chips for e-books or biosensors like blood-sugar monitoring devices. You should even try to get involved in stages like foundry-level fabrication and IP creation. In fact, these kinds of involvements can build a future generation of entrepreneurs in this field.

So don't panic over the current job scenario; the time is perfect to resurrect your thought process on a very large scale. Devote your time to become an expert in VLSI. It will definitely bring 'very large' opportunities in the future. ●

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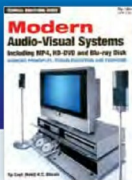
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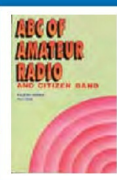
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